

Siyang Ni

Ph.D. Candidate, Criminology & Social Data Analytics

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RESEARCH PROFILE

I am a life-course delinquency and prevention researcher. I develop and apply longitudinal models, quasi-experimental approaches, and machine learning methods to study development from childhood through adolescence and how it affects juvenile delinquency and substance use. I also build open-source analytic tools that improve the accessibility and reproducibility of advanced longitudinal methods for researchers.

EDUCATION

Ph.D., Criminology; Dual-Title Ph.D., Social Data Analytics

Expected 2027

Pennsylvania State University, University Park, PA • ABD

M.Phil., Sociology

Nov 2021

The Chinese University of Hong Kong, Shatin, Hong Kong

Thesis: *Social Strain and Left-behind Children's Delinquency: An Empirical Test of General Strain Theory in Rural China under the Internal Migration Context*

Full funding, Hong Kong Research Grants Council (\$27,740 annually)

B.A., Planning & Public Policy, Summa Cum Laude

May 2016

Rutgers University, New Brunswick, NJ

Honors Thesis: *The Ecological Significance of Hispanic Immigrants and Crime Nexus in Urban Neighborhoods: A Mixed Method Case Study in New Brunswick*

SAS Dean's List, three semesters

DISSERTATION

The Shape of Self-Control: Developmental Course, Family Origins, and Adult Justice-System Contact in the UK Millennium Cohort Study

Committee: Jeremy Staff (Chair) • Derek Kreager • Sarah Brothers • Pui-Wa Lei (External)

One argument tested in three roles: how self-control problems develop, where that development comes from, and what it produces. UK Millennium Cohort Study, approximately 5,600 children followed from age 5 to 23.

Structural equation modeling; latent growth curve, within-family, and mixture models.

Developmental course	“One Construct, Two Timetables: Divergent Development of Hyperactivity–Inattention and Conduct Problems as Components of Self-Control From Age 5 to 17.” The two problems usually lumped together as self-control develop on different schedules: hyperactivity and attention problems improve about three times faster after age 11 than before it, while conduct problems keep improving at the same slow rate they held in childhood. Analysis complete; in writing.
Family origins	“Who Shapes Whom? Family Origins of the Level and Course of Self-Control and Child-to-Parent Effects on Closeness.” A family’s circumstances fix how many self-control problems a child shows by age 5, but from then on the influence runs the “wrong” way: it is the child who increasingly shapes the parent–child relationship, not the parent shaping the child. Analysis complete.
Adult justice contact	“Course, Not Type: Self-Control Development From Childhood to Adolescence and Criminal Justice Contact at Age 23.” How a child’s self-control changed across childhood and adolescence predicts whether they have formal contact with the criminal justice system by age 23, and there is no hidden “type” of child behind that prediction. The risk is a matter of degree, not kind. Analysis complete.

MANUSCRIPTS

UNDER REVIEW

Ni, Siyang. “Is Adolescent Nicotine Vaping Increasingly a Marker of Broader Substance Use? A Temporal Block-Shapley Analysis of US 12th-Grade Students, 2017–2024.” Under review, *Nicotine & Tobacco Research*.

As adolescent vaping declined, it became increasingly differentiated by concurrent non-tobacco substance use rather than by cigarette smoking. Screening and prevention should assess polysubstance use, not smoking alone.

Staff, Jeremy, Eric P. Baumer, and Siyang Ni. “Adolescent Cigarette Smoking Before and After the Proliferation of Electronic Nicotine Delivery Systems: A Period-Based Matching Design.” Under review, *Nicotine & Tobacco Research*.

After matching, past-month smoking probability was 4.3% among post-ENDS youth versus 17.9% among pre-ENDS youth, with a sixteen-fold spread across ENDS-use subgroups.

Ni, Siyang. “When One Migrant Parent Matters More Than Two: Parental Migration, Social Strain, and Delinquency among Adolescents in Rural China.” Under review, *Journal of Research in Crime and Delinquency*.

Adolescents with one migrant parent, not two, reported higher expected delinquency than those with no migrant parent. Migration status conditions which strains link to delinquency.

IN PREPARATION – in addition to the three dissertation papers above

Ni, Siyang. “Exploring the Gendered Association Between Social Strain, Strain-Induced Negative Emotions, and Delinquency among Left-Behind Children in Rural China: A Test of General Strain Theory.” Analysis complete; in writing.

INVITED TALKS

“Using LLMs to Make Data Products.”	Nov 2026
Digital Liberal Arts Lecture Series, Pennsylvania State University · forthcoming	
“A New Machine-Learning Based Framework to Study Repeated Measures in Cross-Sectional Data.”	Oct 2026
Social Data Analytics guest lecture, Pennsylvania State University · forthcoming	
“Development and Heterogeneity of Self-Control.”	Jan 2026
Criminal Justice Research Center, Pennsylvania State University	

CONFERENCE PRESENTATIONS

“Is Adolescent Nicotine Vaping Increasingly a Marker of Broader Substance Use? A Temporal Block-Shapley Analysis of US 12th-Grade Students, 2017–2024.”	Sep 2026
Society for Research on Nicotine and Tobacco – Europe Annual Conference · Oral presentation	
“Three Developmental Trajectories of Self-Control.”	May 2026
Population Association of America Annual Conference · Poster	
“Risk Factors for Adolescent Nicotine Vaping from 2017 to 2023: A Data-Driven Approach.”	Apr 2025
Consortium on Substance Use and Addiction Annual Conference · Poster	
“Revisiting Self-Control Theory: A Longitudinal Analysis Using the UK Millennium Cohort Study.”	Apr 2024
Society for Research on Adolescence Annual Meeting · Poster	
“Exploring the Gendered Association Between Social Strain, Strain-Induced Negative Emotions, and Delinquency Among Left-Behind Children in Rural China: A Test of General Strain Theory.”	Nov 2023
American Society of Criminology Annual Meeting · Panel paper	

“Does Association with Friends Who Use Drugs Increase the Risk of Adolescent Drug Use: Testing Gendered Differential Association Theory.” Consortium on Substance Use and Addiction Annual Conference · Panel paper	Apr 2023
“Parental Migration and Left-Behind Children's Delinquency: A Test of General Strain Theory in Rural China.” Academy of Criminal Justice Sciences Annual Meeting · Panel paper	Mar 2023

GRANTS & AWARDS

RESEARCH FUNDING

Penn State C-SODA Seed Grant, Student PI — \$1,000 Principal Investigator. “Development of Self-Control and Delinquency in Youth: A Longitudinal Test of Self-Control Theory.”	Dec 2025
Penn State RGSO Dissertation Support Award — \$2,000	Nov 2025
Criminal Justice Research Center Seed Grant — \$2,000 Sole author. “Revisiting Self-Control Theory: A Longitudinal Analysis Using the UK Millennium Cohort Study.”	May 2025
Research Enhancement Award — \$800 Sole author. “Revisiting Self-Control Theory: A Longitudinal Analysis Using the UK Millennium Cohort Study.”	Apr 2024

AWARDS & SCHOLARSHIPS

University of Michigan – NSF EITM Scholarship — \$7,425 For attendance at the ICPSR Summer Program in Quantitative Methods.	Jul 2025
Penn State Graduate Scholar Award — \$10,000	Sep 2022
Penn State Startup Scholarship — \$2,500	Sep 2022
Hong Kong Research Grants Council Studentship — \$27,740 annually	2019–2021

Competitive funding awarded to date: \$25,725, plus a fully funded M.Phil. studentship.

SOFTWARE & RESEARCH EXPERIENCE

Open-Source Developer — <i>surveymixr</i> (R package) - Developed an open-source R package integrating growth mixture modeling with complex survey designs (weights, PSUs, strata). - Implemented BLRT, multiple random starts, convergence diagnostics, and the R3STEP distal outcome workflow. - Initial CRAN submission Dec 2025. Repository: github.com/siyangni/surveymixr	Ongoing
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Collaborating Researcher — “Cyberbullying on Reddit: Enforcing Norms and Gaining Upvotes”

Ongoing

With Riley Tucker. Funded by the Penn State Social Science Research Institute.

Data Engineer, Research Support

May–Aug 2025

Office of Digital Pedagogies and Initiatives, Pennsylvania State University

- Built a modular R Shiny dashboard covering 63 years of publishing data.
- Deployed for ongoing use by the Penn State College of the Liberal Arts. [View dashboard](#)

TECHNICAL SKILLS

Methods	Longitudinal and dynamic models · structural equation modeling · growth mixture and latent class models · multilevel models · machine learning
Programming	R · Python · HTML
Statistical software	Stata · Mplus
Data & other	PostgreSQL · L ^A T _E X

TEACHING

INSTRUCTOR OF RECORD

SOC470: Intermediate Social Statistics · Pennsylvania State University

Fall 2025

TEACHING ASSISTANT, PENNSYLVANIA STATE UNIVERSITY

CRIM490: Crime Policy · undergraduate

Fall 2023, Spring 2024

SOC575: Statistical Models for Nonexperimental Research · graduate

Spring 2023

TUTOR, THE CHINESE UNIVERSITY OF HONG KONG

Policing and Society · Sociology of Crime and Deviance · Social Research Methods · Social Problems and Social Policy

SERVICE & PROFESSIONAL MEMBERSHIPS

Department	Planning Committee, Department of Sociology and Criminology, Pennsylvania State University
Memberships	American Society of Criminology · Society for Research on Adolescence · Population Association of America

REFERENCES

All at Pennsylvania State University.

Jeremy Staff, Professor of Sociology, Criminology, and Demography · **Dissertation Chair**
jus25@psu.edu · 814-865-4560

Derek Kreager, Professor of Criminology and Sociology
dak27@psu.edu · 814-867-0217

Sarah Brothers, Assistant Professor of Sociology and Public Policy
sarah.brothers@psu.edu

Riley Tucker, Assistant Professor of Criminology
rqt5439@psu.edu